

Blind Monkey Jam Band

Challenge 4 — Reanalysis triage

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The problem

- A clinical scientist, the morning after the reanalysis cycle finishes, with time for ~20 of 300 unresolved cases before MDT.
 - Today: sort by oldest case, or eyeball candidate counts. Slow, and it misses the cases where the evidence actually moved.
- New ClinVar calls, promoted candidates and lab notes (“parents now sequenced”) sit in different columns — nobody reads all 300.*
- “Which of these 300 should I open first — and why?”

How we addressed it

- A transparent triage: every case gets a score, a tier and a one-line reason. No model, no black box.
- Weights stated up front: ClinVar P/LP upgrade 40 · new high-priority candidate 25 · gene evidence 12 · new HPO term 6 · raw new candidate 3.

Recency dampener ($\times 0.2 < 3 \text{ mo} \dots \times 0.8 < 1 \text{ y}$) · +4/yr since last analysis (cap 20) · incomplete prior run +10–20 · note events +10–30. Case table in → ranking + HTML view out.

- On purpose: no fitting to outcomes. A fresh-context agent built the ranking without ever seeing the outcomes file; we sealed it and opened it only after the ranking was frozen.

No rows dropped for missing values; already-solved cases skipped, not scored.

The result

- Top 20 by triage: 13 solved + 4 follow-up. Oldest-first: 3 solved, 2 slots wasted on already-solved cases. Random: ~2.

Ranks 1–8 all solved. “Review now” (46 cases) holds 21 of 29 solved; “Defer” (170) holds 2.

- Real: triage.py ranks all 300; HTML review page with tier filters and reasons; evaluation vs baseline.
- Not real: no live Talos/ClinVar feeds — signals are the synthetic case table’s columns.

#	Case	Tier	Score	Years since last	Signals	Why this rank
1	CASE_029	Review now	157.8	3.7	ClinVar P/LP 2 new high-priority 1 gene evidence 2	2 ClinVar upgrade(s) to P/LP; 1 new high-priority candidate(s); 2 candidate(s) with strengthened gene-disease evidence; 2 other ClinVar reclassification(s); last analysed 3.7 y ago
2	CASE_185	Review now	126.5	4.1	ClinVar P/LP 1 new high-priority 1 new cand. 4 prior incomplete: qc_fail	1 ClinVar upgrade(s) to P/LP; 1 new high-priority candidate(s); 3 other ClinVar reclassification(s); 4 new candidate(s); prior analysis incomplete (qc_fail); last analysed 4.1 y ago; phenotype_changes=NA treated as 0
3	CASE_293	Review now	117.0	6.5	ClinVar P/LP 1 gene evidence 3 new HPO 2	1 ClinVar upgrade(s) to P/LP; 3 candidate(s) with strengthened gene-disease evidence; 1 other ClinVar reclassification(s); 2 new HPO term(s); last analysed 6.5 y ago
4	CASE_215	Review now	113.0	6.4	ClinVar P/LP 1 new cand. 1 new high-priority 1 new HPO 1	1 ClinVar upgrade(s) to P/LP; 1 new high-priority candidate(s); 3 other ClinVar reclassification(s); 1 new HPO term(s); 1 new candidate(s); last analysed 6.4 y ago

Limits, and working with Claude

- It orders the queue; it does not interpret variants. A scientist still reviews every case — 3 of our top 20 were “no change” despite textbook evidence.
- Claude’s first data inspection printed outcome aggregates — leakage. Caught by asking “can you seal that file?”; fixed with a blind fresh-context build and a written disclosure.

Also: a “second affected sibling” note was under-weighted (solved case at rank 108); recency dampener too harsh at 9–11 months (solved cases at ranks 75, 117).

- Next: tune those two weights, add “what changed since last review” diffs, run on real Talos output across cycles.